

Lecture- 24

Mathematics 1 (15B11MA111)

**Module: Vector Differential
and Integral Calculus**

CO [C105.4]

Topics to be covered

- **Stokes's Theorem**

- ❖ **Definition**

- ❖ **Examples**

- ❖ **Practice Problems**

- **Reference for the lecture**

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Stokes's Theorem

Definition Let S be a piecewise smooth orientable surface bounded by a piecewise smooth simple closed curve C .

Let $v(x, y, z) = v_1(x, y, z)i + v_2(x, y, z)j + v_3(x, y, z)k$ be a vector function which is continuous and has continuous first order partial derivatives in a domain which contains S . If C is traversed in the positive direction, then

$$\oint_C v \cdot dr = \oint_C (v \cdot T) ds = \iint_S (\nabla \times v) \cdot n \, dA$$

Where n is the unit normal to S in the direction of orientation of C .

T is unit tangent vector to C and is given by

$$T = \frac{dx}{ds}i + \frac{dy}{ds}j$$

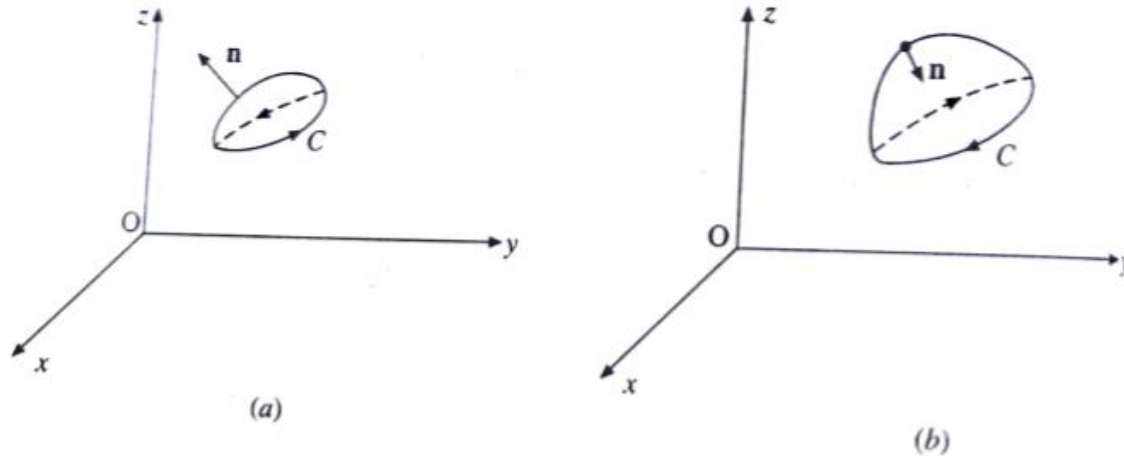


Figure: Directions of C in Stokes's theorem

Remark 1: In terms of components of v , Stoke's theorem can be written as

$$\oint_C [v_1(x, y, z)dx + v_2(x, y, z)dy + v_3(x, y, z)dz] = \iint_S (\nabla \times v) \cdot n \, dA$$

Remark 2: Stokes's theorem states that the value of the surface integral is same for any surface as long as the bounding curve, bounding the projection R on any coordinate plane, is the same curve C . Hence, in the degenerate case, when S coincides with R , we can take $n=k$ or j or i depending on whether the projection is taken on the x - y plane or x - z plane or y - z plane.

Example: Verify Stokes's theorem for the vector field

$$v = (3x - y)i - 2yz^2j - 2zy^2k,$$

where S is the surface of the sphere $x^2 + y^2 + z^2 = 16, z > 0$

Solution: The Projection of S on x-y plane is the circular region

$x^2 + y^2 \leq 16, z = 0$ and the bounding curve C is the circle

$$x^2 + y^2 = 16, z = 0.$$

$$\oint_C v \cdot dr = \oint_C (3x - y)dx - 2yz^2dy - 2zy^2dz = \oint_C (3x - y)dx$$

Since $z=0$. Setting $x = 4\cos\theta, y = 4\sin\theta$

$$\oint_C (3x - y)dx = \int_0^{2\pi} 4(3\cos\theta - \sin\theta)(-4\sin\theta)d\theta = 16\pi$$

Now

$$\nabla \times v = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x - y & -2yz^2 & -2zy^2 \end{vmatrix}$$

$$n = \frac{2xi + 2yj + 2zk}{2\sqrt{x^2 + y^2 + z^2}} = \frac{1}{4}(xi + yj + zk)$$

$$(\nabla \times v) \cdot n = \frac{z}{4}$$

$$\text{Since } dA = \frac{dx \, dy}{n \cdot k}$$

$$\iint_S (\nabla \times v) \cdot n \, dA = \iint_S \frac{z}{4} \, dA = \iint_S \frac{z}{4} \frac{dx \, dy}{n \cdot k} = 16\pi$$

Hence stokes's theorem is proved.

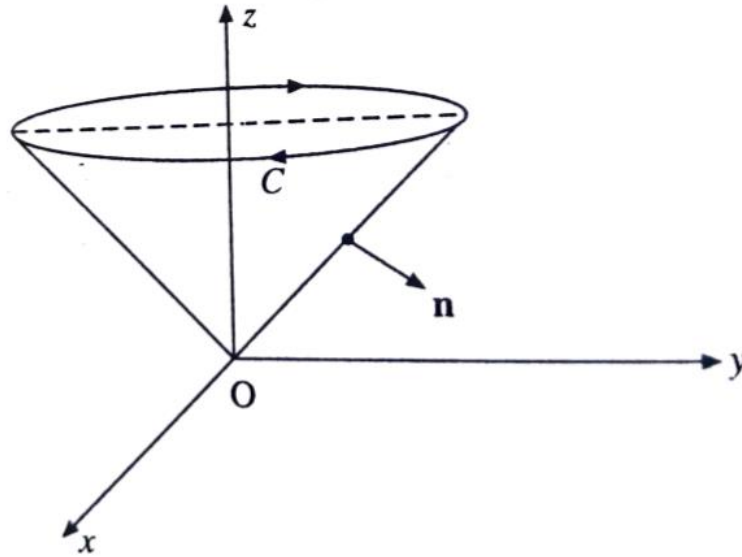
Example: Evaluate $\oint_C 2y^3 dx + x^3 dy + z dz$ where C is the trace of the cone $z = \sqrt{x^2 + y^2}$ intersected by the plane $z=4$ and S is the surface of the cone below $z=4$.

Solution: we have $v = 2y^3 i + x^3 j + z k$ and

$$\text{curl } v = \nabla \times v = \begin{vmatrix} i & j & k \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2y^3 & x^3 & z \end{vmatrix}$$

$$= i(0) + j(0) + k(3x^2 - 6y^2)$$

if the outward normal to S is taken, then it points downwards. Then, the orientation of C is taken as in figure. Alternatively, if the inward normal to S is taken, then C is oriented in the counter clockwise direction.



Let $f(x, y, z) = \sqrt{x^2 + y^2} - z = 0$ be taken as the equation of surface. Then, the normal and unit normal are given by

$$N = \frac{xi + yj}{\sqrt{x^2 + y^2}} - k = \frac{xi + yj - zk}{z}$$

and

$$n = \frac{xi+yj}{\sqrt{x^2+y^2}} - k$$

$$n = \frac{(xi + yj - zk)/z}{\sqrt{(x^2 + y^2 + z^2)}/z^2} = \frac{xi + yj - zk}{\sqrt{2} z}$$

Except at the origin.

$$\begin{aligned} \iint_S (\nabla \times v) \cdot n \, dA &= \iint_S -\frac{(3x^2 - 6y^2)}{\sqrt{2}} \, dA \\ &= - \iint_R -\frac{(3x^2 - 6y^2)}{\sqrt{2}} \frac{dx \, dy}{(-\frac{1}{\sqrt{2}})} \end{aligned}$$

Substituting $x = r\cos\theta, y = r\sin\theta$, we get

$$\begin{aligned}
 \iint_S (\nabla \times v) \cdot n \, dA &= \iint_R (3x^2 - 6y^2) \, dx \, dy \\
 &= \int_{r=0}^4 \int_{2\pi}^0 (3\cos\theta^2 - 6\sin\theta^2) r^3 \, dr \, d\theta \\
 &= 192\pi
 \end{aligned}$$

The bounding curve C is given by $z = 4$, $x^2 + y^2 = 16$.

Now substituting $x = 4\cos\theta$, $y = 4\sin\theta$, we get

$$\begin{aligned}
 \oint_C 2y^3 dx + x^3 dy + z dz &= \oint_C 2y^3 dx + x^3 dy \\
 \oint_C 2y^3 dx + x^3 dy &= \int_{2\pi}^0 64[2\sin\theta^3(-4\sin\theta) + \cos\theta^3(4\cos\theta)] d\theta \\
 &= 192\pi
 \end{aligned}$$

Practice Problems

Q.1: Use Stoke's theorem to evaluate

$$\int_C [(x + 2y)dx + (x - z)dy + (y - z)dz]$$

where C is boundary of the triangle with vertices (2, 0, 0), (0, 3, 0) and (0, 0, 6) oriented in anticlockwise direction. Ans: 15.

Q.2: Verify Stoke's theorem, when $F = yi + zj + xk$ and surface S is part of sphere $x^2 + y^2 + z^2 = 1$ above xy-plane.

Ques 1) $\int_C (x+2y)dx + (x-z)dy + (y-z)dz$

$$\vec{V} = (x+2y)\hat{i} + (x-z)\hat{j} + (y-z)\hat{k}$$

$$\text{Curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+2y & x-z & y-z \end{vmatrix}$$

$$\hat{i}(1+1) - \hat{j}(0-0) + \hat{k}(1-2)$$

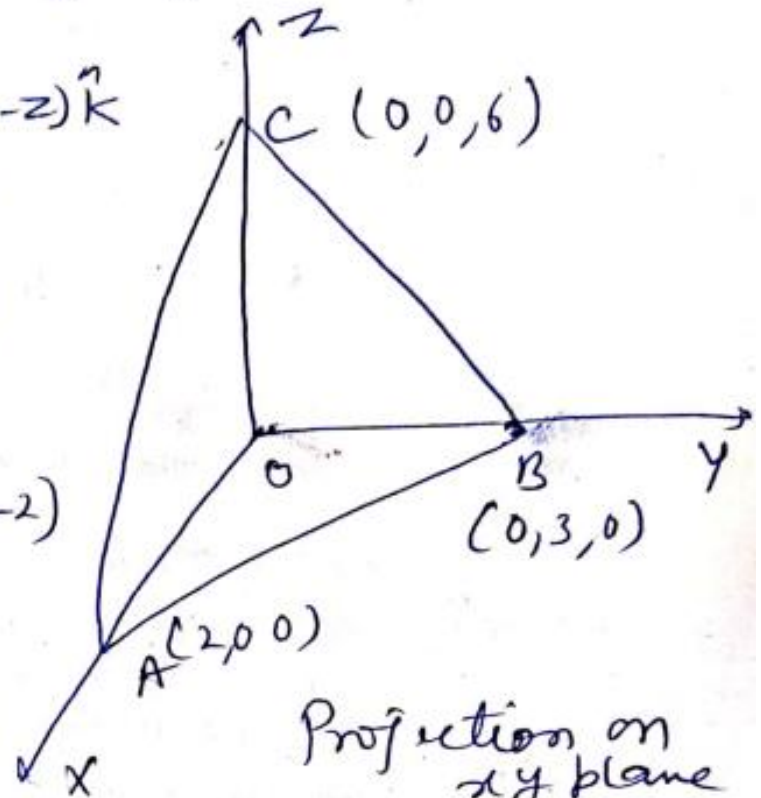
$$= 2\hat{i} - \hat{k}$$

Ans $f(x,y,z) = \frac{x}{2} + \frac{y}{3} + \frac{z}{6} = 1$

$$f(x,y,z) = 3x + 2y + z = 6$$

$$\hat{n} = \frac{3\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{14}}$$

$$ds = \frac{dx dy}{|\hat{n} \cdot \hat{k}|}$$



Projection on
xy plane

$$\begin{aligned}
 \iint_S \text{curl } \vec{V} \cdot \hat{n} \, ds &= \iint_S (2\hat{i} - \hat{k}) \cdot \left(\frac{3\hat{i} + 2\hat{j} + \hat{k}}{\sqrt{14}} \right) \cdot \frac{dx dy}{\sqrt{14}} \\
 &= \iint_S 5 \, dx dy = 5 \times \iint_{\Delta AOB} dx dy \quad \text{Area of } \Delta AOB \\
 &= 5 \times \frac{1}{2} \times 2 \times 3 = 15
 \end{aligned}$$

Ques 2) $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$
 $x^2 + y^2 + z^2 = 1$

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

where C is the unit circle $x^2 + y^2 = 1, z = 0$

$$\vec{F} \cdot d\vec{r} = y \, dx + z \, dy + x \, dz$$

on unit circle $z = 0; dz = 0$

$$\oint_C \vec{F} \cdot d\vec{r} = \int_C y dx \quad \text{radius} = 1$$

$$\text{Let } x = \cos \theta, \quad y = \sin \theta$$

$$dx = -\sin \theta d\theta$$

$$= \int_0^{2\pi} \sin \theta (-\sin \theta) d\theta = - \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= -\pi$$

$$\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix} = -\hat{i} - \hat{j} - \hat{k}$$

$$\hat{n} = \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2\sqrt{x^2 + y^2 + z^2}} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{\sqrt{x^2 + y^2 + z^2}}$$

$$\text{Curl } \vec{F} \cdot \hat{n} = (-x - y - z)$$

$$\text{Projection on } xy \text{ plane } ds = \frac{dx dy}{\sqrt{2}}$$

$$\text{curl } \vec{F} \cdot \hat{n} = (-x - y - z)$$

Projection on xy plane $ds = \frac{dxdy}{z}$

$$\iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = \iint_S (-x - y - z) \frac{dxdy}{z}$$

$$= - \iint_S \left(\frac{x}{z} + \frac{y}{z} + 1 \right) dxdy$$

$$z = \sqrt{1 - x^2 - y^2}$$

$$= - \int_{-1}^1 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left(\frac{x}{\sqrt{1-x^2-y^2}} + \frac{y}{\sqrt{1-x^2-y^2}} + 1 \right) dx dy$$

$$= -2 \int_{-1}^1 \int_0^{\sqrt{1-x^2}} \left(\frac{x}{\sqrt{1-x^2-y^2}} + 1 \right) dy dx$$

$$= -2 \int_{-1}^1 \left[x \left[\sin^{-1} \left(\frac{y}{\sqrt{1-x^2}} \right) \right]_0^{\sqrt{1-x^2}} + y \right]_0^{\sqrt{1-x^2}} dx$$

$$= -2 \int_{-1}^1 \left(\frac{\pi}{2} x + \sqrt{1-x^2} \right) dx$$

$$= -4 \int_0^1 \sqrt{1-x^2} dx$$

$$= -4 \left[\frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1$$

$$= -4 \times \frac{\pi}{4} = -\pi.$$

Thank You